

The empirical recurrence for OEIS A235834: recovering a conjecture that is not written down

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Abstract

OEIS A235834 records “Empirical recurrence of order 84”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 150$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235834 is “Number of $(2+1)X(n+1)$ 0..2 arrays with the difference of the upper median and minimum value of each $2X2$ subblock in lexicographically nondecreasing order columnwise and nonincreasing rowwise.”. It has offset 1 and begins

549, 8160, 112146, 1403872, 16791083, 193298612, 2172780930, 23997087333, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:51:59, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 226$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 150$ of the 226, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 150$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 452$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	26	c_{22}	−245433919829914801	c_{43}	12768699935070806218378511	c_{64}	−13
c_2	−32	c_{23}	1057873855732745636	c_{44}	−3670933302411848787193453	c_{65}	−6
c_3	−4454	c_{24}	2148436973789488968	c_{45}	−30558240547199410343706042	c_{66}	79
c_4	28087	c_{25}	−10587278593353441059	c_{46}	11270807931592645205420400	c_{67}	27
c_5	328317	c_{26}	−15815405046206562500	c_{47}	63978334971012082226531327	c_{68}	−3
c_6	−3279408	c_{27}	89720976206583596758	c_{48}	−28340771723774361521328086	c_{69}	−8
c_7	−12945559	c_{28}	97883076051602168617	c_{49}	−117020323001842354090834756	c_{70}	15
c_8	208045978	c_{29}	−649874405556722568702	c_{50}	59903604913907718772963720	c_{71}	17
c_9	231073694	c_{30}	−506317596694191054139	c_{51}	186584922325702446492781504	c_{72}	−4
c_{10}	−8822693775	c_{31}	4051849183305193753768	c_{52}	−107742287261328361992115860	c_{73}	−
c_{11}	3554642104	c_{32}	2153447247437225214984	c_{53}	−258558526534447438916262676	c_{74}	13
c_{12}	273276381279	c_{33}	−21862601393681003151741	c_{54}	165835133106885428655233856	c_{75}	−
c_{13}	−378079590666	c_{34}	−7228968192286741283701	c_{55}	310123945905405348776260200	c_{76}	−
c_{14}	−6498338885115	c_{35}	102507190784108097123310	c_{56}	−218854214494959547839615568	c_{77}	2
c_{15}	13702781382559	c_{36}	16889494830085201577473	c_{57}	−320237339372662797841735216	c_{78}	2
c_{16}	122475652630388	c_{37}	−418928958412082716667914	c_{58}	247478882471814613386051392	c_{79}	−
c_{17}	−331203315742272	c_{38}	−10642708016272781421673	c_{59}	282686283629410093461960224	c_{80}	−
c_{18}	−1869983368077691	c_{39}	1495580208225318653622731	c_{60}	−239140234356513279422023808	c_{81}	−
c_{19}	6058196727508725	c_{40}	−139901630366463108182987	c_{61}	−211330033133279527031975552	c_{82}	−
c_{20}	23490168522258645	c_{41}	−4670642956260075263808998	c_{62}	196563276414819747294701824	c_{83}	−
c_{21}	−88365832253284379	c_{42}	918556367983333592852843	c_{63}	132090638424043982312622080	c_{84}	−

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84 + 4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

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